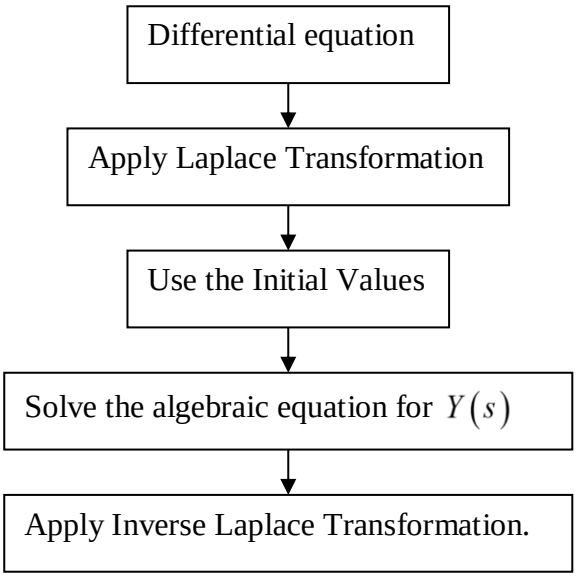
Lecture 3
Application Of Laplace Transformation

Solving Differential Equations

The Laplace transformation is useful in solving differential equations. There are four steps to follow, such as



Important formulae

$$\mathcal{L}\{f'(t)\} = \mathcal{L}\left\{\frac{df(t)}{dt}\right\} = sF(s) - f(0).$$

$\mathcal{L}\{f''(t)\} = \mathcal{L}\left\{\frac{d^2f(t)}{dt^2}\right\} = s^2F(s) - sf(0) - f'(0)$ where $f(0)$, and $f'(0)$are the initial values of f and f'.

$$\mathcal{L}\{f'''(t)\} = \mathcal{L}\left\{\frac{d^3f(t)}{dt^3}\right\} = s^3F(s) - s^2f(0) - sf'(0) - f''(0).$$

The general case for the Laplace transforms of an nth derivative is

$$\mathcal{L}\{f^n(t)\} = \mathcal{L}\left\{\frac{d^nf(t)}{dt^n}\right\} = s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots\cdots - f^{(n-1)}(0).$$

Solving Ordinary Differential equations with constant coefficients:

The Laplace transform is useful in solving linear ordinary differential equations with constant coefficients. Having obtained expressions for the Laplace transforms of derivatives, we are now able to use Laplace transform, and inverse Laplace transform methods to solve ordinary differential equations with constant coefficients. To illustrate this, consider the general second-order differential equation

$$\frac{d^2y}{dt^2} + \alpha\frac{dy}{dt} + \beta y = f(t) \quad \text{or} \quad y''(t) + \alpha y'(t) + \beta y(t) = f(t) \qquad (1)$$

Where α and β are constants, subject to initial conditions

$$y(0) = A, y'(0) = B \qquad (2)$$

where A and B are given constants. On taking the Laplace transform of both sides and using condition (2), we obtain the algebraic equation for determination of $\mathcal{L}\{y(t)\} = Y(s)$. The required solution is then obtained by finding the inverse Laplace transform of $Y(s)$. The method is easily extended for the higher order differential equations.

Example 3.1

Solve the differential equation or the initial value problem

$y''(t) + y(t) = e^t, \quad y(0) = 1, \quad y'(0) = -2.$

Solution:

Given,

$y''(t) + y(t) = e^t,$

$\Rightarrow \mathcal{L}\{y''(t)\} + \mathcal{L}\{y(t)\} = \mathcal{L}\{e^t\}$ [applying Laplace transformation]

$\Rightarrow s^2Y(s) - sy(0) - y'(0) + Y(s) = \frac{1}{s-1}$ [let, $\mathcal{L}\{y(t)\} = Y(s)$]

$\Rightarrow s^2Y(s) - s(1) - (-2) + Y(s) = \frac{1}{s-1}$ [using the initial values]

$\Rightarrow Y(s) = \frac{s^2-3s+3}{(s^2+1)\,(s-1)}$ [solving the equation]

$\Rightarrow \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{A}{s-1} + \frac{Bs+C}{s^2+1}\right\}$ [applying inverse Laplace transformation and using partial fraction]

$$\frac{s^2 - 3s + 3}{(s^2+1)\,(s-1)} \equiv \frac{A}{s-1} + \frac{Bs+C}{s^2+1}$$

$\Rightarrow s^2 - 3s + 3 \equiv A(s^2+1) + (Bs+C)(s-1)$

$\Rightarrow s^2 - 3s + 3 \equiv (A+B)s^2 + (C-B)s + A - C$

Equating coefficients

$$A + B = 1$$
$$C - B = -3$$
$$A - C = 3$$

Solving we get, $A = \frac{1}{2},\ B = \frac{1}{2}, C = -\frac{5}{2}$

$$\Rightarrow \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{1/2}{s-1} + \frac{s/2}{s^2+1} - \frac{5/2}{s^2+1}\right\}$$

$\Rightarrow y(t) = \frac{e^t}{2} + \frac{1}{2}\cos t - \frac{5}{2}\sin t.$

Therefore, the solution of the differential equation is $y(t) = \frac{e^t}{2} + \frac{1}{2}\cos t - \frac{5}{2}\sin t.$

Example 3.2

A resistance $\boldsymbol{R = 3\Omega}$ in series with inductance L= $\mathbf{1}\,\boldsymbol{Henry}$ relates to e.m.f $\boldsymbol{E(t) = 9t}$. The equation of current $\boldsymbol{i(t)}$ is given by

$$L\frac{di}{dt} + Ri = 9t$$

Assuming initial current is zero, use Laplace transform to find the current $i(t)$.

Solution:

Given,

$$L\frac{di}{dt} + Ri = 9t$$
$$\Rightarrow \frac{di}{dt} + 3i = 9t$$

$\mathcal{L}\left\{\frac{di}{dt}\right\} + 3\mathcal{L}\{i\} = 9\mathcal{L}\{t\}$ [applying Laplace transformation]

$\Rightarrow sI(s) - i(0) + 3I(s) = \frac{9}{s^2}$ [let, $\mathcal{L}\{i(t)\} = I(s)$]

$\Rightarrow sI(s) - 0 + 3I(s) = \frac{9}{s^2}$ [using the initial values]

$\Rightarrow I(s) = \frac{9}{(s+3)s^2}$ [Solving the equation]

$\Rightarrow \mathcal{L}^{-1}\{I(s)\} = \mathcal{L}^{-1}\left\{\frac{9}{(s+3)s^2}\right\}$ [applying inverse Laplace transformation]

$\Rightarrow i(t) = \mathcal{L}^{-1}\left\{\frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+3}\right\}$[using partial fraction]

$$\frac{9}{(s+3)s^2} \equiv \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+3}$$

$$\Rightarrow 9 \equiv As(s+3) + B(s+3) + Cs^2$$

$$\Rightarrow 9 \equiv (A+C)s^2 + (3A+B)s + 3B$$

Equating coefficients

$$A + C = 0$$
$$3A + B = 0$$
$$3B = 9$$

Solving we get, $A = -1$, $B = 3$, $C = 1$

Hence,

$i(t) = \mathcal{L}^{-1}\left\{\frac{-1}{s} + \frac{3}{s^2} + \frac{1}{s+3}\right\}$

$\Rightarrow i(t) = -\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + 3\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s+3}\right\}$

$\Rightarrow i(t) = -1 + 3t + e^{-3t}$.

Example 3.3

An inductor of 2 henrys, a resistor of 16 ohms and a capacitor of .02 farads are connected in series with an e.m.f. of E volts. At $t = 0$ the charge on the capacitor and current in the circuit are zero. Find the charge and current at any time $t > 0$ if E= 300 (volts).

Let $q(t)$ and $i(t)$ be the instantaneous charge and current respectively at time t. By Kirchhoff's law's, we have

$$2\frac{di}{dt} + 16i + \frac{q}{0.02} = E$$

$\Rightarrow 2\frac{d^2q}{dt^2} + 16\frac{dq}{dt} + 50q = E$ [since $i = \frac{dq}{dt}$]

$\Rightarrow 2\frac{d^2q}{dt^2} + 16\frac{dq}{dt} + 50q = E$…… (1)

With the initial conditions

$$q(0) = 0, i(0) = \dot{q}(0) = 0.$$

Solution:

If $E = 300$, then equation (1) becomes

$$\frac{d^2q}{dt^2} + 8\frac{dq}{dt} + 25q = 150$$

$\Rightarrow \mathcal{L}\left\{\frac{d^2q}{dt^2}\right\} + 8\mathcal{L}\left\{\frac{dq}{dt}\right\} + 25\mathcal{L}\{q\} = \mathcal{L}\{150\}$ [applying Laplace transform]

$\Rightarrow \{s^2Q(s) - s\,q(0) - q'(0)\} + 8\{sQ(s) - q(0)\} + 25Q(s) = \frac{150}{s}$ [let, $\mathcal{L}\{q(t)\} = Q(s)$]

$\Rightarrow \{s^2Q(s) - s\,.0 - 0\} + 8\{sQ(s) - 0\} + 25Q(s) = \frac{150}{s}$ [using the initial values]

$\Rightarrow s^2Q(s) + 8sQ(s) + 25Q(s) = \frac{150}{s}$

$\Rightarrow (s^2 + 8s + 25)Q = \frac{150}{s}$

$\Rightarrow Q(s) = \frac{150}{s(s^2+8s+25)}$ [solving the equation]

$\Rightarrow \mathcal{L}^{-1}\{Q(s)\} = \mathcal{L}^{-1}\left\{\frac{150}{s(s^2 + 8s + 25)}\right\}$

$\Rightarrow q(t) = \mathcal{L}^{-1}\left\{\frac{A}{s} + \frac{Bs+C}{(s^2+8s+25)}\right\}$ [using partial fraction]

$$\frac{150}{s(s^2 + 8s + 25)} \equiv \frac{A}{s} + \frac{Bs + C}{(s^2 + 8s + 25)}$$

$\Rightarrow 150 \equiv A(s^2 + 8s + 25) + (Bs + C)s$

$\Rightarrow 150 \equiv (A + B)s^2 + (8A + C)s + 25A$

Equating coefficients

$$A + B = 0$$
$$8A + C = 0$$
$$25A = 150$$

Solving we get, $A = 6, B = -6, C = -48$

$$q(t) = \mathcal{L}^{-1}\left\{\frac{6}{s} - \frac{6s + 48}{(s^2 + 8s + 25)}\right\}$$

$$\Rightarrow q(t) = \mathcal{L}^{-1}\left\{\frac{6}{s} - \frac{6(s + 4) + 24}{(s + 4)^2 + 9}\right\}$$

$$\Rightarrow q(t) = \mathcal{L}^{-1}\left\{\frac{6}{s}\right\} - \mathcal{L}^{-1}\left\{\frac{6(s + 4)}{(s + 4)^2 + 9}\right\} - \mathcal{L}^{-1}\left\{\frac{24}{(s + 4)^2 + 9}\right\}$$

$q(t) = 6 - 6e^{-4t}Cos\ t - 8\ e^{-4t} \sin 3t$ [applying inverse Laplace transformation]

$i(t) = \frac{dq}{dt} = 50\ e^{-4t} \sin 3t.$

Example 3.4

The current $i(t)$ in an electrical circuit is given by the DE, $\frac{d^2i}{dt^2} + 2\frac{di}{dt} = \begin{cases} 0, & 0 < t < 10 \\ 1, & 10 < t < 20 \\ 0, & t > 20 \end{cases}$

$i(0) = \frac{di}{dt}(0) = 0.$ Determine current $i(t)$.

Solution: Using unit step function the DE becomes,

$\frac{d^2i}{dt^2} + 2\frac{di}{dt} = u(t - 10) - u(t - 20)$ ……(1).

$\Rightarrow \mathcal{L}\left\{\frac{d^2i}{dt^2}\right\} + 2\mathcal{L}\left\{\frac{di}{dt}\right\} = \mathcal{L}\{u(t - 10)\} - \mathcal{L}\{u(t - 20)\}$ [applying Laplace transform]

$\Rightarrow \left\{s^2I(s) - si(0) - \frac{di}{dt}(0)\right\} + 2\{sI(s) - i(0)\} = \frac{e^{-10s}}{s} - \frac{e^{-20s}}{s}$

$[\because \mathcal{L}\{f(t).u(t - a)\} = e^{-as}\mathcal{L}\{f(t + a)\}$

$\Rightarrow I(s)\big(s(s + 2)\big) = \frac{e^{-10s}}{s} - \frac{e^{-20s}}{s}$

$$\Rightarrow I(s) = \frac{1}{s(s+2)}\left(\frac{e^{-10s}}{s} - \frac{e^{-20s}}{s}\right)$$

Applying inverse Laplace transform, we get

$$\mathcal{L}^{-1}\{I(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s(s+2)}\left(\frac{e^{-10s}}{s} - \frac{e^{-20s}}{s}\right)\right\} \dots\dots\dots\dots (2).$$

We know $\mathcal{L}^{-1}\{e^{-as}G(s)\} = u(t-a).g(t-a)$ and $\mathcal{L}^{-1}\left\{\frac{1}{s^2(s+2)}\right\} = \frac{t}{2} + \frac{e^{-2t}}{4} - \frac{1}{4}$.

$$\therefore i(t) = \left[\frac{(t-10)}{2} + \frac{e^{-2(t-10)}}{4} - \frac{1}{4}\right]u(t-10) - \left[\frac{(t-20)}{2} + \frac{e^{-2(t-20)}}{4} - \frac{1}{4}\right]u(t-20).$$

$$= \begin{cases} 0, & 0 < t < 10 \\ \dfrac{(t-10)}{2} + \dfrac{e^{-2(t-10)}}{4} - \dfrac{1}{4}, & 10 < t < 20 \\ 5 + \dfrac{e^{-2(t-10)}}{4} - \dfrac{e^{-2(t-20)}}{4}, & t > 20 \end{cases}.$$

Shifted data problems

This is a short name for initial value problem with initial conditions referring to some later instant $t = t_0$ instead of $t = 0$. In this case, the conditions $y(0)$ and $y'(0)$ occurring in the Laplace transform approach cannot be used immediately.

Example 3.5

$y'' + y = 2t, \quad y\left(\frac{\pi}{4}\right) = \frac{\pi}{2}, \quad y'\left(\frac{\pi}{4}\right) = 2 - \sqrt{2}$.

Solution: set $t = \bar{t} + \frac{\pi}{4}$ so that $t = \frac{\pi}{4}$ gives $\bar{t} = 0$ and then Laplace transform becomes applicable throughout.

Now, the shifted problem is

$\bar{y}'' + \bar{y} = 2\left(\bar{t} + \frac{\pi}{4}\right), \ \bar{y}(0) = \frac{\pi}{2}$ and $\bar{y}'(0) = 2 - \sqrt{2}$.

Using Laplace transform on both sides we obtain

$$s^2\,\bar{Y}(s) - s\bar{y}(0) - \bar{y}'(0) = \frac{2}{s^2} + \frac{\pi}{2}\,\frac{1}{s}$$

$$\Rightarrow \bar{Y}(s) = \frac{2}{s^2\,(s^2+1)} + \frac{\pi}{2}\,\frac{1}{s\,(s^2+1)} + \frac{\pi}{2}\,\frac{s}{s^2+1} + \left(2 - \sqrt{2}\right)\frac{1}{s^2+1}$$

Applying inverse Laplace on both sides,

$$\Rightarrow \bar{y}(\bar{t}) = 2\,(\bar{t} - \sin\bar{t}) + \frac{\pi}{2}\,(1 - \cos\bar{t}) + \frac{\pi}{2}\cos\bar{t} + \left(2 - \sqrt{2}\right)\sin\bar{t}$$

Substituting $\bar{t} = t - \frac{\pi}{4}$ we obtain the solution

$y(t) = 2t - \sin t + \cos t$.

Problem set 3.1

1. **Apply Laplace transforms to solve the following ordinary differential equations and hence justify your answer, where $y' \equiv \frac{dy(t)}{dt}$ and $y'' \equiv \frac{d^2y(t)}{dt^2}$:**

a. $y'(t) = 3;$ $y(0) = 2.$ **Ans:** $y(t) = 3t + 2.$
b. $y'(t) = 4t;$ $y(0) = 1.$ **Ans:** $y(t) = 2\,t^2 + 1.$
c. $y'(t) = 2t - 1;$ $y(0) = 3.$ **Ans:** $y(t) = t\,(t-1) + 3.$
d. $y'(t) = t^2;$ $y(0) = 4.$ **Ans:** $y(t) = \frac{t^3}{3} + 4.$
e. $y'(t) = e^{2t};$ $y(0) = 2.$ **Ans:** $y(t) = \frac{e^{2t}}{2} + \frac{3}{2}.$
f. $y'(t) + y(t) = 2\,;$ $y(0) = 0.$ **Ans:** $y(t) = 2 - 2\,e^{-t}.$
g. $y''(t) = 5\,;$ $y(0) = 1, y'(0) = 2.$ Ans: $y(t) = \frac{t\,(5t+4)}{2} + 1.$
h. $y''(t) - 2\,y'(t) = \cos t\,;$ $y(0) = 0, y'(0) = 1.$
Ans: $f(t) = \frac{7}{10}\,e^{2t} - \frac{2}{5}\left(\sin t + \frac{1}{2}\cos t\right) - \frac{1}{2}.$
i. $y''(t) + 3\,y'(t) - y(t) = e^t;$ $y(0) = y'(0) = 0.$
Ans: $f(t) = \frac{1}{3}\,e^t - \frac{1}{3}\,e^{\frac{-3t}{2}}\left[\cosh\left(\frac{\sqrt{13}}{2}\,t\right) + \frac{5}{\sqrt{13}}\sinh\left(\frac{\sqrt{13}}{2}\,t\right)\right]$
j. $y''(t) - 7y'(t) + 12y(t) = 0, y(0) = 2, y'(0) = 1.$
Ans: $y(t) = -5e^{4t} + 7e^{3t}.$
k. $y''(t) + y(t) = \begin{cases} t, & 0 < t < 1 \\ 0, & t > 1 \end{cases}, y(0) = 0, y'(0) = 0.$
Ans: $y(t) = t - \sin t$ if $0 < t < 1$ and $\cos(t-1) + \sin(t-1) - \sin t$ if $t > 1.$

2. **Problems from book: Solve the IVPs by the Laplace transformation:(** Advanced Engineering Mathematics – Erwin Kreyszig**)**

a. $y'' - y = t;\ y(0) = 1, y'(0) = 1$
b. $y'' + y' + 9y = 0;\ y(0) = 0.16, y'(0) = 0$
c. $y' + y = 5\sin(2t);\ y(0) = 0$
d. $y' + 2y = 0;\ y(0) = 1.5$
e. $y'' - y' - 6y = 0;\ y(0) = 11, y'(0) = 28$
f. $y'' + 9y = 10e^{-t};\ y(0) = 0, y'(0) = 0$
g. $y'' - \frac{1}{4}y = 0;\ y(0) = 12, y'(0) = 0$
h. $y'' - 6y' + 5y = 29\cos(2t);\ y(0) = 3.2, y'(0) = 6.2$
i. $y'' + 7y' + 12y = 21e^{3t};\ y(0) = 3.5, y'(0) = -10$
j. $y'' - 4y' + 4y = 0;\ y(0) = 8.1, y'(0) = 3.9$
k. $y'' - 4y' + 3y = 6t - 8;\ y(0) = 0, y'(0) = 0$
l. $y'' + 3y' + 2.25y = 9t^3 + 64;\ y(0) = 1, y'(0) = 31.5$
m. $y'' + 4y' + 5y = 50t;\ y(0) = 5, y'(0) = -5$
n. $y'' + 16y = 4\delta(t - \pi);\ y(0) = -1, y'(0) = 0$
o. $y'' - y' - 2y = 12u(t - \pi)\sin t\,;\ y(0) = 1, y'(0) = -1$
p. $y'' + 4y = \delta(t - \pi) - \delta(t - 2\pi);\ y(0) = 1, y'(0) = 0$
q. $y'' + 3y' + 2y = 2u(t - 2);\ y(0) = 0, y'(0) = 0$

Solving Simultaneous Ordinary Differential Equations by Laplace Transform

Example 3.6

$$\begin{cases} x'(t) = 2x(t) - 3y(t) \\ y'(t) = y(t) - 2x(t) \end{cases} \text{ subject to } x(0) = 8, y(0) = 3.$$

Solution:

Taking the Laplace transforms of both equations

$\Rightarrow \mathcal{L}\{x'(t)\} = 2\mathcal{L}\{x(t)\} - 3\mathcal{L}\{y(t)\}$

$\mathcal{L}\{y'(t)\} = \mathcal{L}\{y(t)\} - 2\mathcal{L}\{x(t)\}$

$\Rightarrow sX(s) - x(0) = 2X(s) - 3Y(s)$

$sY(s) - y(0) = Y(s) - 2X(s)$

$\Rightarrow (s-2)X(s) + 3Y(s) = 8$

$2X(s) + (s-1)Y(s) = 3$ [using initial condition and rearranging]

Now solving these two equations simultaneously using **Cramer's rule** and partial fraction we get,

$$X(s) = \frac{\begin{vmatrix} 8 & 3 \\ 3 & s-1 \end{vmatrix}}{\begin{vmatrix} s-2 & 3 \\ 2 & s-1 \end{vmatrix}} = \frac{8s-17}{s^2-3s-4} = \frac{8s-17}{(s+1)(s-4)} = \frac{5}{s+1} + \frac{3}{s-4}$$

$$Y(s) = \frac{\begin{vmatrix} s-2 & 8 \\ 2 & 3 \end{vmatrix}}{\begin{vmatrix} s-2 & 3 \\ 2 & s-1 \end{vmatrix}} = \frac{3s-22}{s^2-3s-4} = \frac{3s-22}{(s+1)(s-4)} = \frac{5}{s+1} - \frac{2}{s-4}$$

Now taking inverse Laplace transform we get,

$\mathcal{L}^{-1}\{X(s)\} = \mathcal{L}^{-1}\left\{\frac{5}{s+1} + \frac{3}{s-4}\right\}$

$\mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{5}{s+1} - \frac{2}{s-4}\right\}$

$\Rightarrow x(t) = 5e^{-t} + 3e^{4t}$

$y(t) = 5e^{-t} - 2e^{4t}$

Problem set 3.2

1. **Solve the following system of differential equations where $x(t) \equiv x$, $y(t) \equiv y$, $y' \equiv \frac{dy(t)}{dt}$ and $y' \equiv \frac{dx(t)}{dt}$, using Laplace transformation. Also justify your answers.**

a. $x' = y;\ y' = 16x;\ \ x(0) = 0, y(0) = 4.$
   Answer: $x(t) = sinh\, 4\,t,\ y(t) = 4\, cosh\, 4\,t$
b. $x' = -4y;\ y' = x;\ \ x(0) = 2, y(0) = 0.$
   Answer: $x(t) = 2\, cos\, 2\,t,\ y(t) = sin\, 2\,t$
c. $x' = 2x + y;\ y' = 4x + 2y;\ \ x(0) = 1, y(0) = 6.$
   Answer: $\ x(t) = e^{2t}(cosh\, 2\,t + 3\, sinh\, 2\,t), y(t) = e^{2t}(6\, cosh\, 2\,t + 2\, sinh\, 2\,t).$
d. $x' = 3x + y;\ y' = 4x + 3y;\ \ x(0) = 3, y(0) = 2.$
   Answer: $x(t) = e^{3t}(3\cosh 2t + \sinh 2t),\ \ y(t) = e^{3t}(2\cosh 2t + 6\sinh 2t).$

2. Problems from Book :Using the Laplace transformation, solve the system of ODEs: (Advanced Engineering Mathematics – Erwin Kreyszig**)**

a. $y_1' = -y_1 + 4y_2,\ y_2' = 3y_1 - 2y_2;\ y_1(0) = 3, y_2(0) = 4$

b. $y_1' = 5y_1 + y_2,\ y_2' = y_1 + 5y_2;\ y_1(0) = 1, y_2(0) = -3$

c. $y_1' = 4y_1 + y_2,\ y_2' = -y_1 + 2y_2;\ y_1(0) = 3, y_2(0) = 1$

d. $y_1'' = y_1 + 3y_2,\ y_2'' = 4y_1 - 4e^t;\ y_1(0) = 2, y_1'(0) = 3, y_2(0) = 1, y_2'(0) = 2$

e. $y_1'' = -2y_1 + 2y_2,\ y_2'' = 2y_1 - 5y_2;\ y_1(0) = 1, y_1'(0) = 0, y_2(0) = 3, y_2'(0) = 0$

f. $y_1' = y_2,\ y_2' = -4y_1 + \delta(t - \pi);\ y_1(0) = 0, y_2(0) = 0$

g. $y_1' = 2y_1 - 4y_2,\ y_2' = y_1 - 3y_2;\ y_1(0) = 3, y_2(0) = 0$

h. $y_1' = 2y_1 + 4y_2,\ y_2' = y_1 + 2y_2;\ y_1(0) = -4, y_2(0) = -4$

Application: Formulation of ODE from word problem

General talk:

The Laplace transform is widely used in the following science and engineering field** .

1. Analysis of electronic circuits.
2. System modeling.
3. Digital signal processing.
4. Nuclear physics.
5. Process control.

The following examples highlight the importance of Laplace transformation in different engineering fields.

Example 3.7

The following example is based on the concepts from nuclear physics. Consider the following first order linear differential equation

$$\frac{dN}{dt} = -\lambda N \dots\dots\dots (1)$$

This equation is the fundamental relationship describing radioactive decay, where $N = N(t)$

represents the number of undecayed atoms remaining in a sample of a radioactive isotope at time t and λ is the decay constant.

We can use Laplace transform to solve this equation (1).

Rearranging the above equation (1) we get,

$$\frac{dN}{dt} + \lambda N = 0 \dots\dots\dots (2)$$

Taking Laplace transform on both sides of (2)

$$s\,L(N) - N(0) + \lambda\,L(N) = 0$$

$$\Rightarrow s\,\overline{N} - N_0 + \lambda\,\overline{N} = 0$$

Where $L(N) = \overline{N}$ and $N(0) = N_0$

$$\Rightarrow \overline{N} = \frac{N_0}{s+\lambda}$$

Now taking inverse Laplace transform on both sides we get,

$$N(t) = N_0\, e^{-\lambda t}.$$

Which is indeed the correct form for radioactive decay.

Example 3.8 : Four-Terminal RLC-Network
Find the charge and current if $R = 20\ \Omega$, $L = 1\ H$, $C = 10^{-3}$ F, the input is $\delta(t)$ (a unit impulse at time $t = 0$) current and charge are zero at time $t = 0$

Solution*:* The required model to the above problem is

$$Li' + iR + \frac{q}{C} = Lq'' + Rq' + \frac{q}{C} = \delta(t)$$
$$q'' + 20q' + 1000q = \delta(t)$$

Applying Laplace transform, we get

$$(s^2 + 20s + 1000)Q(s) = 1\ [\because Q(s) = \mathcal{L}\{q\}, \mathcal{L}\{\delta(t-0) = e^{-0s} = 1]$$
$$Q(s) = \frac{1}{(s^2+20s+1000)} = \frac{1}{(s+10)^2+900}$$

By applying first shifting theorem we obtain damped oscillations for q and then find the current using $i(t) = \frac{dq}{dt}$

$$q(t) = \mathcal{L}^{-1}\{Q(s)\} = \frac{1}{30}e^{-10t}\sin 30t$$

and

$$i(t) = \frac{dq}{dt} = -\frac{1}{3}\,e^{-10t}\big(sin(30\,t) - 3\,cos\,(30\,t)\big)$$

Example 3.9 : Unrepeated Complex Factors. Damped Forced Vibrations
Solve the initial value problem for a damped mass-spring system acted upon by a sinusoidal force for some time interval,

$$y'' + 2y' + 2y = r(t),$$

where $r(t) = 10\sin 2t$ if $0 < t < \pi$ and 0 if $t > \pi$,

$y(0) = 1, y'(0) = -5$

Solution. The subsidiary equation

$$(s^2Y - s + 5) + 2(sY - 1) + 2Y = 10\frac{2}{s^2+4}\big(1 - e^{-\pi\vartheta}\big)$$

We collect the $Y(s)$ terms, $(s^2 + 2s + 2)$, take $-s + 5 - 2 = -s + 3$ to the right, and solve,

$$= \frac{20}{(s^2+4)(s^2+2s+2)} - \frac{20e^{-\pi s}}{(s^2+4)(s^2+2s+2)} + \frac{s-3}{s^2+2s+2}.$$

For the last fraction we get from the first shifting theorem

$$\mathcal{L}^{-1}\left\{\frac{s+1-4}{(s+1)^2+1}\right\} = e^{-t}(\cos t - 4\sin t).$$

In the first fraction we have unrepeated complex roots, hence a partial fraction representation

$$\frac{20}{(s^2+4)(s^2+2s+2)} = \frac{As+B}{s^2+4} + \frac{Ms+N}{s^2+2s+2}.$$

Multiplication by the common denominator gives

$$20 = (As+B)(s^2+2s+2) + (Ms+M)(s^2+4)$$

We determine A, B, M, N. Equating the coefficients of each power of s on both sides gives the four equations
(a) $[s^3]$: $0 = A + M$
(b) $[s^2]$: $0 = 2A + B + N$
(c) $[s]$: $0 = 2A + 2B + 4M$
(d) $[s^0]$: $20 = 2B + 4N$.

We can solve this, for instance, obtaining $M = -A$ from (a), then $A = B$ from (c), then $N = -3A$ from (b), and finally $A = -2$ from (d). Hence $A = -2, B = -2, M = 2, N = 6$, and the first fraction has the representation

$$\frac{-2s-2}{s^2+4} + \frac{2(s+1)+6-2}{(s+1)^2+1}$$

Inverse transform: $-2\cos 2t - \sin 2t + e^{-t}(2\cos t + 4\sin t)$.

The sum of this inverse is the solution of the problem for $0 < t < \pi$, namely (the sines cancel),

$$y(t) = 3e^{-t}\cos t - 2\cos 2t - \sin 2t \text{ if } 0 < t < \pi$$

In the second fraction, taken with the minus sign, we have the factor $e^{-\pi s}$, so that from and the second shifting theorem we get the inverse transform of this fraction for $t > 0$ in the form

$$+2\cos(2t-2\pi) + \sin(2t-2\pi) - e^{-(t-\pi)}[2\cos(t-\pi) + 4\sin(t-\pi)]$$
$$= 2\cos 2t + \sin 2t + e^{-(t-\pi)}(2\cos t + 4\sin t)$$

The sum of all three inverses is the solution for $t > \pi$,

$$y(t) = e^{-t}[(3+2e^{\pi})\cos t + 4e^{\pi}\sin t] \text{ if } t > \pi$$

Figure 136 shows (9) (for $0 < t < \pi$) and (10) (for $t > \pi$), a beginning vibration, which goes to zero rapidly because of the damping and the absence of a driving force after $t = \pi$.

Problem set 3.3

RLC circuit: $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = v(t)$

1. Using the Laplace transform and showing the details, find the charge $q(t)$ and the current $i(t)$ in the circuit , assuming zero initial current and charge with

$$R = 2\Omega, L = 1H, C = 0.5F, v(t) = 1kV \text{ if } 0 < t < 2 \text{ and } 0 \text{ if } t > 2.$$

Ans:

$$q(t) = \begin{cases} 500[1 - e^{-t}(\cos t + \sin t)], & 0 \le t < 2 \\ 500[e^{-(t-2)}(\cos(t-2) + \sin(t-2)) - e^{-t}(\cos t + \sin t)] & t > 2 \end{cases}$$

and

$$i(t) = \begin{cases} 1000e^{-t}\sin t, & 0 \le t < 2 \\ 1000e^{-t}\sin t - 1000e^{-(t-2)}\sin(t-2), & t > 2 \end{cases}$$

2. Using the Laplace transform and showing the details, find the charge $q(t)$ and the current $i(t)$ in the circuit , assuming zero initial current and charge with

$$R = 6\,\Omega, \mathrm{L} = 1\mathrm{H}, \mathrm{C} = 0.125\ \mathrm{F}, \mathrm{v(t)} = 4\mathrm{e}^{-2\mathrm{t}}\ \mathrm{Volt}$$

RC circuit

3. Using the Laplace transform and showing the details, find the charge $q(t)$ and the current $i(t)$ in the circuit, assuming zero initial charge with

$$R = 3\Omega, C = 0.5F, v(t) = 4\mathrm{t}$$

RL circuit

4. Using the Laplace transform and showing the details, find the charge $q(t)$ and the current $i(t)$ in the circuit, assuming zero initial current and charge with

$$L = 1H, R = 3\Omega, v(t) = 2\mathrm{e}^{-\mathrm{t}}$$

5. Using the Laplace transform and showing the details, find the current $i(t)$ in the circuit, assuming initial current, $i(0) = 1A$

$$L = 1H, R = 2\Omega, v(t) = 4$$

LC circuit

6. Using the Laplace transform and showing the details, find the charge $q(t)$ and the current $i(t)$ in the circuit, assuming zero initial current and charge with

$$L = 1H, \mathrm{C} = \frac{1}{9}F, v(t) = 4\sin(\mathrm{t})$$

Mass-Spring system:

$$my''(t) + by'(t) + ky(t) = F_{ext}$$

Or

$$y''(t) + \frac{b}{m}y'(t) + \frac{k}{m}y(t) = \frac{F_{ext}}{m}$$

Undamped Oscillation: $k > 0, b = 0, F_{ext} = 0$

7. Solve the initial value problem for an undamped mass-spring system,
$$y'' + 4y = 0,$$

where $y(0) = 1$, $y'(0) = 2$

Damped Oscillation:

Underdamped: $\boldsymbol{b^2 - 4mk < 0}$

8. Solve the initial value problem using Laplace Transformation for an underdamped mass-spring system given by,
$$y'' + 2y' + 2y = 0,$$

where $y(0) = 1$, $y'(0) = 0$

9. Solve the initial value problem for an underdamped mass-spring system,

$$y'' + 2y' + 5y = 0,$$

where $y(0) = 1$, $y'(0) = 1$

10. Solve the initial value problem using Laplace Transformation for a underdamped mass-spring system given by,
$$y'' + 4y' + 5y = 0,$$

where $y(0) = 1$, $y'(0) = -1$

Critically damped: $\boldsymbol{b^2 - 4mk = 0}$

11. Solve the initial value problem using Laplace Transformation for a critically damped mass-spring system given by,
$$y'' + 2y' + y = 0,$$

where $y(0) = 1$, $y'(0) = 0$

12. Solve the initial value problem using Laplace Transformation for a critically damped mass-spring system given by,
$$y'' + 6y' + 9y = 0,$$

where $y(0) = 1$, $y'(0) = -2$

Overdamped: $b^2 - 4mk > 0$

13. Solve the initial value problem using Laplace Transformation for a overdamped mass-spring system given by,

$$y'' + 4y' + 3y = 0,$$

where $y(0) = 1, y'(0) = -4$

14. Solve the initial value problem using Laplace Transformation for a overdamped mass-spring system given by,

$$y'' + 18y' + 56y = 0,$$

where $y(0) = 2, y'(0) = -18$

External force: $F_{ext} \neq 0$

15. Solve the initial value problem using Laplace Transformation for a mass-spring system given by,

$$y'' + 3y' + 2y = e^{-t},$$

where $y(0) = 1, y'(0) = -1$